

Furkan Güzelant

MSc Student in Computer Science | Bilkent University, Ankara, Turkey

furkan.guzelant@bilkent.edu.tr | Personal Website | LinkedIn | GitHub

Education

Bilkent University, , M.Sc. in Computer Science — Sept 2024 – Present

- **GPA**: 3.90/4.0
- Under the supervision of Asst. Prof. Ayşegül Dündar at the Generative Deep Learning Research Lab.
- Granted a **Department Scholarship** at the time of enrollment
- **Research interests:** 3D Generative Models, Diffusion Models, 3D Stylization

École Polytechnique Fédérale de Lausanne (EPFL), Erasmus Exchange — Feb 2023 – June 2023

- **GPA**: 5.42/6.0
- **Coursework**: Deep Learning for Autonomous Vehicles, Computer Graphics

Bilkent University, , B.Sc. in Computer Science — Sept 2019 – June 2024

- **GPA**: 3.94/4.0
- **Full Scholarship**: Ranked 300th among over 2.5 million students in the university entrance exam
- Graduated with Summa Cum Laude honors
- **Selected Coursework:** Machine Learning, Deep Generative Networks, Computer Graphics, Computer Networks

Publications

Continuous Angular Control in Text-to-Image Generation via Geometric Token Encoding — Oct 2026

Furkan Güzelant, Basavaraj Sunagad, Ayşegül Dündar, Adam Kortylewski

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2027

StyleFusion360: View-Consistent Head Stylization via Adaptive Style Modulation — June 2026

Furkan Güzelant, Arda Göktoğan, Tarık Kaya, Ayşegül Dündar

European Conference on Computer Vision (ECCV), 2026

Identity Preserving 3D Head Stylization with Multiview Score Distillation — June 2025

Bahri Batuhan Bilecen, Ahmet Berke Gökmen, ***Furkan Güzelant***, Ayşegül Dündar

IEEE/CVF International Conference on Computer Vision (ICCV), 2025

Geometry-Aware 3D Scene Completion via Differentiable Next-View Selection — 2026

Arda Göktoğan, ***Furkan Güzelant***, Duygu Ceylan, Ayşegül Dündar

Manuscript in submission

Experience

Research Collaboration, Remote — Feb 2026 – Present

- Conducted collaborative research with Adam Kortylewski on precise continuous geometric control in text-to-image diffusion models using periodic, role-specific geometric token encoding.
- Introduced Stochastic Localization Relaxation (SLR) and preference-based optimization to improve spatial grounding and semantic-role consistency.
- Resulted in a first-author paper accepted to WACV 2027.

Teaching Assistant, Bilkent University — Sep 2024 – Present

- Course assisted: CS421 Computer Networks

Candidate Engineer, InterLabs – Ankara, Turkey — June 2023 – Sep 2024

- Developed and trained machine learning models for accurate detection and tracking of small, fast-moving objects
- Built a robust Automatic Speech Recognition (ASR) and Speaker Diarization pipeline to transcribe and distinguish Turkish meeting recordings
- Fine-tuned large language models (LLMs) for Turkish dialogue summarization and implemented Retrieval-Augmented Generation (RAG) systems to enhance search and information retrieval

Test Engineering Intern, HAVELSAN – Ankara, Turkey — Aug 2022 – Sep 2022

- Developed automation testing using Selenium and TestNG

Machine Learning Engineer Intern, DataBoss Analytics & Security – Ankara, Turkey — July 2022 – Aug 2022

- Completed a time-series forecasting project using data analysis and feature engineering.
- Implemented supervised machine learning models such as XGBoost and LightGBM to make predictions from historical data, gaining hands-on experience with Python, scikit-learn, and NumPy.

Projects

Bird's Eye View (BEV) Semantic Segmentation

- Designed and developed a solution for the Bird's Eye View (BEV) Semantic Segmentation task using PanopticBEV as the baseline model.
- Conducted experiments to improve the accuracy of the model

Retrieval-Augmented Generation (RAG) for Image Synthesis

- Enhanced a RAG-based image generation pipeline by improving retrieval accuracy and incorporating iterative refinement.

Yoldas - A blind-assistance application

- Developed a mobile app to support visually impaired individuals in leading more independent lives
- Enabled real-time video connections with volunteers for assistance with daily tasks and emotional support
- Integrated AI features for image and document recognition, along with an accessible UI and smart feedback mechanisms
- Designed and implemented the frontend and core AI functionalities of the application
- Project recognized with the **Social Impact Award** at the Bilkent CS Fair 2024
- **Technologies used:** Flutter, Python

Technical Skills

Programming Languages: Python, Java, JavaScript, C/C++, SQL

Application Frameworks: Java Spring Boot, Flutter, React

AI/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, Pandas, Numpy, Matplotlib

Developer Tools: Docker, LaTeX, Git, GitHub, Jira

Languages: English (Proficient), German (Beginner)

Achievements

Summa Cum Laude Honors: Ranked 4th out of 178 students

National University Entrance Exam: Ranked 300th among 2.5 million candidates

IELTS Academic Overall Band Score: 8.0

Academic Personnel and Postgraduate Education Entrance Exam: Ranked 39th out of 128 thousand applicants

The Scientific and Technological Research Council of Türkiye (TUBITAK) BIDEB Scholarship